

Universal Calculus

The Kleban Operator and Branched Equalizer Geometry

A formal, readable operator calculus for latent-to-realspace reconstruction

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Formal calculus draft

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Primitive terminal equation

$$\mathbb{K}_{\theta,\beta} = \delta_A \circ \mathcal{R}_\theta^\beta$$

$$\text{Real}_0^\beta = \ker \mathbb{K}_{\theta,\beta}$$

$$\text{Real}_\varepsilon^\beta = \mathbb{K}_{\theta,\beta}^{-1}(B_\varepsilon(0))$$

The Kleban Operator is the primitive obstruction-to-realspace operator. Its zero locus is exact realspace, its finite ball preimage is finite-resolution realspace, its derivative induces defect-normal geometry, and its action norm drives dynamics.

Condensed from the branched tri-cylindrical local realspace construction. The calculus is formal; empirical correspondence requires additional observables and interpretation maps.

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1 Purpose

This document rewrites the branched tri-cylindrical construction as a compact calculus. The goal is not to add new physical claims. The goal is to isolate the smallest formal operator grammar that reproduces the construction and makes its symmetries explicit.

Central compression

The construction is generated by one operator:

$$\mathbb{K}_{\theta,\beta} : \Lambda \longrightarrow V^3, \quad \mathbb{K}_{\theta,\beta}(\lambda) = (\mathcal{R}_0^\beta(\lambda) - \mathcal{R}_1^\beta(\lambda), \mathcal{R}_1^\beta(\lambda) - \mathcal{R}_2^\beta(\lambda), \mathcal{R}_2^\beta(\lambda) - \mathcal{R}_0^\beta(\lambda)).$$

This is the **Kleban Operator**. It measures the failure of three branch-selected projection sectors to agree. Everything else is derived from it:

realspace	=	$\ker \mathbb{K},$
finite realspace	=	$\mathbb{K}^{-1}(B_\varepsilon(0)),$
defect	=	$\mathbb{K}(\lambda),$
linearized defect	=	$d\mathbb{K},$
defect-normal metric	=	$(d\mathbb{K})^* \kappa(d\mathbb{K}),$
dynamics	=	$\text{Crit}(\ \mathbb{K}\ ^2 + \text{regularity} + \text{boundary} + \text{branch variation}).$

Here “universal” means universal inside the specified formal class: stable comparison substrate, one global phase, three axis sectors, branch-selected transverse forms, axis-coboundary compatibility, and variational selection. It does not mean that every physical system is claimed to realize the calculus.

2 Objects and Notation

The calculus uses a small number of objects. They are separated into substrate, branch, projection, defect, and variational layers.

Symbol	Meaning
$I \subseteq \mathbb{N}$	discrete index set for chart states or history steps
S_n	raw state set at chart n
T_n	admissible transformation family at chart n
Λ_n	latent comparison space generated by stable comparisons
$A = \{0, 1, 2\}$	axis index set, read as $0 = x, 1 = y, 2 = z$
$V = \mathbb{R}^3$	target real coordinate vector space
$B = \{E, P, H\}$	branch set: elliptic cylinder, polar parent, hyperbolic cylinder
θ	single global phase variable
α_i	local phase offset in axis sector i
a_i, r_i	axial coordinate and radial scale in sector i
$\beta_i : \Lambda \rightarrow B$	branch field for sector i
$\mathcal{R}_i^\beta$	branch-selected reconstruction map for sector i
δ_A	axis-coboundary operator
$\mathbb{K}_{\theta, \beta}$	Kleban Operator, the primitive defect operator
κ	contrast form on defect variations
$\mathcal{S}$	action functional selecting admissible histories

When no index n is shown, the formula applies at one chart. A history is obtained by applying the same definitions at each $n \in I$.

3 Substrate Layer: Stable Comparison

Definition 3.1 (Atomic comparison). An atomic comparison is a tuple

$$C = (a, b, \delta, \chi),$$

where $a, b \in S$ are compared states, δ is a contrast value, and χ is a context locator.

Definition 3.2 (Stable comparison). Given tolerance $\varepsilon \geq 0$ and an admissible transformation family T , define

$$\text{Stab}_\varepsilon(S, T) = \{(a, b, \delta, \chi) : |\delta(a, b; \chi) - \delta(Ta, Tb; T\chi)| \leq \varepsilon \quad \forall T \in T\}.$$

Only stable comparisons are allowed to generate latent structure.

Definition 3.3 (Latentization). The latent comparison space is the structured quotient generated by stable comparisons:

$$\mathcal{L}_\varepsilon(S, T) = \Lambda := \langle \text{Stab}_\varepsilon(S, T) \rangle / \sim.$$

The relation $\sim$ identifies comparison data that are indistinguishable under the allowed stability tests.

Substrate rule

Raw states do not define space. A raw datum becomes admissible only after it survives stable comparison. Therefore the calculus starts with relational invariance, not with a pre-existing manifold.

4 Phase and Branch Layer

4.1 One global phase

There is exactly one global phase variable θ . At a discrete chart index,

$$\theta_n = \theta_0 + \omega n.$$

In a differentiable approximation,

$$\theta(t) = \theta_0 + \omega t.$$

Each projection sector reads the same phase through a sector offset:

$$\psi_i = \theta + \alpha_i.$$

Elliptic and polar branches read ψ_i modulo 2π . Hyperbolic branches read the real lift of ψ_i .

4.2 Branch trigonometry

Let

$$B = \{E, P, H\}.$$

The branch signs are

$$\varepsilon_b = \begin{cases} +1, & b = E, P, \\ -1, & b = H. \end{cases}$$

Define branch cosine and branch sine by

$$C_b(\psi) = \begin{cases} \cos \psi, & b = E, P, \\ \cosh \psi, & b = H, \end{cases} \quad S_b(\psi) = \begin{cases} \sin \psi, & b = E, P, \\ \sinh \psi, & b = H. \end{cases}$$

Definition 4.1 (Universal transverse branch map). For branch $b \in B$, radius or scale $\rho \geq 0$, phase readout ψ , and hyperbolic sheet $\sigma \in \{+1, -1\}$, define

$$\Gamma_b(\rho, \psi, \sigma) = \begin{cases} \rho(C_b(\psi), S_b(\psi)), & b = E, P, \\ \sigma \rho(C_b(\psi), S_b(\psi)), & b = H. \end{cases}$$

The sheet variable is inactive for E and P .

Proposition 4.1 (Branch quadratic identity). Let

$$q_b(u, v) = u^2 + \varepsilon_b v^2.$$

Then

$$q_b(\Gamma_b(\rho, \psi, \sigma)) = \rho^2.$$

Proof. For E and P , this is $\rho^2(\cos^2 \psi + \sin^2 \psi) = \rho^2$. For H , this is $\rho^2(\cosh^2 \psi - \sinh^2 \psi) = \rho^2$. The sheet sign squares away. $\square$

Branch interpretation

E = compact elliptic fixed-radius cylinder,
 P = polar parent with unfrozen radius,
 H = hyperbolic outward cylinder with transverse signature flip.

The ordinary cylinder is the fixed-radius leaf of the polar parent.

5 Projection Layer

5.1 Axis insertion maps

Let $A = \{0, 1, 2\}$ with coordinate basis (e_0, e_1, e_2) , read as (x, y, z) . Define transverse insertion maps $J_i : \mathbb{R}^2 \rightarrow V$ by

$$J_0(u, v) = (0, u, v), \quad J_1(u, v) = (v, 0, u), \quad J_2(u, v) = (u, v, 0).$$

Each J_i inserts a transverse pair into the coordinate plane normal to axis i .

For each sector i , define latent coordinate functions

$$a_i : \Lambda \rightarrow \mathbb{R}, \quad r_i : \Lambda \rightarrow \mathbb{R}_{\geq 0}, \quad \alpha_i : \Lambda \rightarrow \mathbb{R}.$$

Let $R_i > 0$ denote a fixed elliptic radius. Define branch radius

$$\rho_i^b(\lambda) = \begin{cases} R_i, & b = E, \\ r_i(\lambda), & b = P, H. \end{cases}$$

Definition 5.1 (Branch-selected sector projection). For $i \in A$ and branch field $\beta_i : \Lambda \rightarrow B$, define

$$\mathcal{R}_i^\beta(\lambda) = a_i(\lambda)e_i + J_i \Gamma_{\beta_i(\lambda)}(\rho_i^{\beta_i(\lambda)}(\lambda), \theta + \alpha_i(\lambda), \sigma_i).$$

The full projection triple is

$$\mathcal{R}_\theta^\beta(\lambda) = (\mathcal{R}_0^\beta(\lambda), \mathcal{R}_1^\beta(\lambda), \mathcal{R}_2^\beta(\lambda)) \in V^3.$$

Written out for $i = 0$,

$$\begin{aligned} \mathcal{R}_0^E(\lambda) &= (a_0, R_0 \cos \psi_0, R_0 \sin \psi_0), \\ \mathcal{R}_0^P(\lambda) &= (a_0, r_0 \cos \psi_0, r_0 \sin \psi_0), \\ \mathcal{R}_0^H(\lambda) &= (a_0, \sigma_0 r_0 \cosh \psi_0, \sigma_0 r_0 \sinh \psi_0). \end{aligned}$$

The other axes are produced by J_1 and J_2 , not by new definitions.

6 The Kleban Operator

6.1 Axis coboundary

Define the oriented axis-coboundary operator

$$\delta_A : V^3 \longrightarrow V^3$$

by

$$\delta_A(X_0, X_1, X_2) = (X_0 - X_1, X_1 - X_2, X_2 - X_0).$$

It records pairwise failure of cyclic compatibility.

Definition 6.1 (The Kleban Operator). The **Kleban Operator** is

$$\mathbb{K}_{\theta, \beta} := \delta_A \circ \mathcal{R}_\theta^\beta$$

or componentwise

$$\mathbb{K}_{\theta, \beta}(\lambda) = (\mathcal{R}_0^\beta(\lambda) - \mathcal{R}_1^\beta(\lambda), \mathcal{R}_1^\beta(\lambda) - \mathcal{R}_2^\beta(\lambda), \mathcal{R}_2^\beta(\lambda) - \mathcal{R}_0^\beta(\lambda)).$$

It maps a latent comparison point to its branch-selected gluing defect.

Definition 6.2 (Raw Kleban Operator). When one wants the operator to act from raw substrate data, define

$$\widehat{\mathbb{K}}_{\varepsilon, \theta, \beta} := \mathbb{K}_{\theta, \beta} \circ \mathcal{L}_\varepsilon.$$

Thus

$$S \xrightarrow{\mathcal{L}_\varepsilon} \Lambda \xrightarrow{\mathcal{R}_\theta^\beta} V^3 \xrightarrow{\delta_A} V^3.$$

The latent form $\mathbb{K}_{\theta, \beta}$ is the clean primitive once Λ has been constructed.

Fundamental semantic object

Kleban Operator = axis coboundary of branch-selected projections.

It is simultaneously an obstruction, a defect field, an equalizer generator, and the seed of induced geometry.

7 Realspace as Kernel and Equalizer

Definition 7.1 (Exact realspace). Exact branch-realspace is the kernel of the Kleban Operator:

$$\text{Real}_0^\beta = \ker \mathbb{K}_{\theta, \beta} = \{\lambda \in \Lambda : \mathbb{K}_{\theta, \beta}(\lambda) = 0\}.$$

Equivalently,

$$\text{Real}_0^\beta = \text{Eq}(\mathcal{R}_0^\beta, \mathcal{R}_1^\beta, \mathcal{R}_2^\beta).$$

Definition 7.2 (Finite-resolution realspace). For tolerance $\varepsilon \geq 0$,

$$\text{Real}_\varepsilon^\beta = \mathbb{K}_{\theta, \beta}^{-1}(B_\varepsilon(0)) = \{\lambda \in \Lambda : \|\mathbb{K}_{\theta, \beta}(\lambda)\| \leq \varepsilon\}.$$

Definition 7.3 (Realspace image). On Real_0^β , all sector projections agree. Therefore the image projection

$$\pi^\beta : \text{Real}_0^\beta \rightarrow V$$

is well-defined by

$$\pi^\beta(\lambda) = \mathcal{R}_i^\beta(\lambda)$$

for any $i \in A$. The equality condition makes the choice of i irrelevant.

Proposition 7.1 (Kernel-equalizer equivalence).

$$\lambda \in \ker \mathbb{K}_{\theta, \beta} \iff \mathcal{R}_0^\beta(\lambda) = \mathcal{R}_1^\beta(\lambda) = \mathcal{R}_2^\beta(\lambda).$$

Proof. If the three projections agree, every cyclic difference is zero, so $\mathbb{K} = 0$. Conversely, if $\mathbb{K} = 0$, then $\mathcal{R}_0 = \mathcal{R}_1$, $\mathcal{R}_1 = \mathcal{R}_2$, and $\mathcal{R}_2 = \mathcal{R}_0$, hence the three projections are equal. $\square$

8 Differential Kleban Calculus

8.1 Linearized Kleban Operator

At a point $\lambda \in \Lambda$, define

$$D\mathbb{K}_\lambda = d\mathbb{K}_{\theta,\beta}|_\lambda : T_\lambda\Lambda \rightarrow T_{\mathbb{K}(\lambda)}V^3.$$

This is the linearized defect map.

Since $\mathbb{K} = \delta_A \circ \mathcal{R}_\theta^\beta$, the chain rule gives

$$D\mathbb{K}_\lambda = \delta_A \circ d\mathcal{R}_\theta^\beta|_\lambda.$$

Thus the same coboundary operator controls finite defect and infinitesimal defect.

8.2 Kleban-normal metric

Let κ_λ be a symmetric bilinear contrast form on defect variations. Define the quadratic normal contrast

$$Q_{\mathbb{K},\lambda}(v) = \kappa_\lambda(D\mathbb{K}_\lambda v, D\mathbb{K}_\lambda v).$$

Polarization gives

$$g_{\mathbb{K},\lambda}^N(u, v) = \frac{1}{2} [Q_{\mathbb{K},\lambda}(u + v) - Q_{\mathbb{K},\lambda}(u) - Q_{\mathbb{K},\lambda}(v)]$$

and equivalently

$$g_{\mathbb{K}}^N = (D\mathbb{K})^* \kappa (D\mathbb{K}).$$

Remark 8.1 (Metric meaning). $g_{\mathbb{K}}^N$ is a defect-normal metric. It measures how strongly a latent tangent direction changes the compatibility defect. If $v \in \ker D\mathbb{K}_\lambda$, then $g_{\mathbb{K},\lambda}^N(v, w) = 0$ for all w when κ is nondegenerate on the image. Such a direction may be gauge, physical tangent, or inadmissible degeneracy. This classification is not automatic.

8.3 Tangential realspace metric

If an intrinsic metric on the realspace image is needed, introduce a target contrast form h on V and define

$$g_{\mathbb{K}}^T = (d\pi^\beta)^* h (d\pi^\beta).$$

The calculus therefore separates normal defect stiffness from tangential realspace geometry:

$$g_{\mathbb{K}}^N \neq g_{\mathbb{K}}^T \quad \text{unless an additional identification rule is supplied.}$$

9 Gluing Connection and Curvature

The phase offsets α_i may vary over Λ . Define sector gluing one-forms

$$\mathcal{A}_i = d\alpha_i + \omega_i, \quad i \in A,$$

where ω_i represents additional local transport in sector i .

Define sector curvatures

$$\mathcal{F}_i = d\mathcal{A}_i.$$

The relative curvature is the axis coboundary of the curvature triple:

$$\Omega_{\mathbb{K}} := \delta_A \mathcal{F} = (\mathcal{F}_0 - \mathcal{F}_1, \mathcal{F}_1 - \mathcal{F}_2, \mathcal{F}_2 - \mathcal{F}_0).$$

Definition 9.1 (Flat Kleban gluing). A branch sector has flat relative gluing when

$$\Omega_{\mathbb{K}} = 0.$$

It has nontrivial relative gluing curvature when $\Omega_{\mathbb{K}} \neq 0$.

Curvature rule

The same axis-coboundary δ_A controls both the finite projection defect and relative curvature:

$$\mathbb{K} = \delta_A \mathcal{R}_{\theta}^{\beta}, \quad \Omega_{\mathbb{K}} = \delta_A d(d\alpha + \omega).$$

10 Branch Admissibility

A branch field

$$\beta = (\beta_0, \beta_1, \beta_2) : \Lambda \rightarrow B^3$$

is discrete. If it is allowed to flicker freely, the system can suppress defect by branch noise rather than coherent gluing. The calculus therefore requires a branch admissibility law.

Axiom 10.1 (Branch stability). The branch field is locally stable on open compatible regions. It may change only across persistent defect interfaces, prescribed boundary strata, or explicitly admitted transition sets.

A compact smooth approximation is

$$\text{TV}(\beta) < \infty$$

with support constraint

$$\text{supp}(d\beta) \subseteq \{\lambda : \|\mathbb{K}_{\theta, \beta}(\lambda)\| > \eta\} \cup \partial\Lambda \cup \Sigma_{\text{prescribed}}.$$

Here $\eta \geq 0$ is a persistence threshold and $\Sigma_{\text{prescribed}}$ is a declared set of admissible branch strata.

Definition 10.1 (Branch transition cost). A branch transition cost is a functional

$$J_{\beta} \in [0, \infty]$$

that penalizes spatial, latent, or temporal variation in β . In a smooth approximation one may take $J_{\beta} = \text{TV}(\beta)$. In a discrete implementation one may count weighted branch changes between neighboring comparison cells.

11 Dynamics

A local history is a sequence

$$H = (X_n, \theta_n, \beta_n, \mathcal{A}_n, \mathbb{K}_n, \kappa_n)_{n \in I}.$$

The universal action has the form

$$\mathcal{S}[H] = \sum_{n \in I} \left[\|\mathbb{K}_{\theta_n, \beta_n}\|_{\kappa_n}^2 + \lambda \mathcal{R}_n + \mu \mathcal{B}_n + \nu J_{\beta, n} \right].$$

Here:

$$\begin{aligned} \|\mathbb{K}\|_{\kappa}^2 &= \text{gluing-defect cost,} \\ \mathcal{R}_n &= \text{regularity cost,} \\ \mathcal{B}_n &= \text{boundary cost,} \\ J_{\beta, n} &= \text{branch-transition cost,} \\ \lambda, \mu, \nu \geq 0 &= \text{weights.} \end{aligned}$$

Definition 11.1 (Valid history). A valid history is stationary or locally minimizing:

$$H^* \in \text{Crit}(\mathcal{S}).$$

In a discrete implementation,

$$X_{n+1} \in \arg \min_{X' \in \text{Adm}(X_n)} \left[\|\mathbb{K}_{\theta', \beta'}(X')\|_{\kappa_n}^2 + \lambda \mathcal{R}(X') + \mu \mathcal{B}(X') + \nu J_{\beta'}(X') \right].$$

Dynamical interpretation

The system evolves by reducing accumulated Kleban defect while preserving comparison stability, one-phase coherence, branch admissibility, and boundary constraints. Persistent defect is not automatically error; it can become structure when it is stable under admissible minimization.

12 Gauge and Symmetry

12.1 Global phase gauge

The phase read by sector i is

$$\psi_i = \theta + \alpha_i.$$

Therefore the transformation

$$\theta \mapsto \theta + c, \quad \alpha_i \mapsto \alpha_i - c$$

leaves every ψ_i invariant and hence leaves $\mathbb{K}$ invariant.

12.2 Local connection gauge

The one-form

$$\mathcal{A}_i = d\alpha_i + \omega_i$$

is preserved by

$$\alpha_i \mapsto \alpha_i + \eta_i, \quad \omega_i \mapsto \omega_i - d\eta_i.$$

Thus $\mathcal{F}_i = d\mathcal{A}_i$ and $\Omega_{\mathbb{K}} = \delta_A \mathcal{F}$ are gauge-invariant under this compensation.

12.3 Axis permutation symmetry

Let $\pi \in S_3$ act by permuting axes. If $J_i, a_i, r_i, \alpha_i, R_i, \sigma_i$, and β_i are permuted coherently, then

$$\mathbb{K}_{\theta, \pi\beta}(\pi\lambda) = \pi\mathbb{K}_{\theta, \beta}(\lambda).$$

Consequently,

$$\text{Real}_\varepsilon^{\pi\beta} = \pi\text{Real}_\varepsilon^\beta$$

and, if $\kappa, \mathcal{R}, \mathcal{B}$, and J_β are axis-symmetric,

$$\mathcal{S}[\pi H] = \mathcal{S}[H].$$

Symmetry group

The natural symmetry envelope is

$$S_3 \ltimes G_\theta \ltimes G_{\mathcal{A}},$$

where S_3 permutes axes, G_θ shifts global phase with compensating offsets, and $G_{\mathcal{A}}$ shifts local offsets with compensating transport forms.

13 Complex Target and Real Forms

The target may be taken as $V = \mathbb{R}^3$ or as a real slice of $\mathbb{C}^3$. In the complexified version, first define

$$\widehat{\mathcal{R}}_i^\beta : \Lambda \rightarrow \mathbb{C}^3.$$

A real sector is then chosen by an antilinear involution

$$\tau : \mathbb{C}^3 \rightarrow \mathbb{C}^3, \quad \tau^2 = \text{id},$$

with reality condition

$$\tau(\widehat{\mathcal{R}}_i^\beta) = \widehat{\mathcal{R}}_i^\beta.$$

Elliptic and hyperbolic branches are different real forms of the complexified transverse quadratic relation. At the complexified level,

$$\psi \mapsto i\psi$$

interchanges circular and hyperbolic functions up to the selected real slice. The real branch selection determines which effective geometry is admitted.

14 Axioms of Universal Calculus

Axiom 14.1 (Stable latent substrate). Only stable comparisons generate latent structure:

$$\Lambda = \mathcal{L}_\varepsilon(S, T) = \langle \text{Stab}_\varepsilon(S, T) \rangle / \sim.$$

Axiom 14.2 (Unique phase). A local universe has exactly one global phase θ . Every sector reads θ through $\psi_i = \theta + \alpha_i$.

Axiom 14.3 (Branch triform). Each sector selects a branch in $B = \{E, P, H\}$ and all branches are realized by one transverse map Γ_b satisfying $q_b(\Gamma_b) = \rho^2$.

Axiom 14.4 (Kleban compatibility). Effective realspace is not primitive. It is the compatibility kernel of the Kleban Operator:

$$\text{Real}_0^\beta = \ker \mathbb{K}_{\theta, \beta}.$$

Axiom 14.5 (Metric derived from defect or image). A metric is not primitive. Defect-normal geometry is induced by $(d\mathbb{K})^* \kappa(d\mathbb{K})$; tangential image geometry requires an independent target contrast h and the pullback $(d\pi)^* h(d\pi)$.

Axiom 14.6 (Branch admissibility). The discrete branch field cannot flicker arbitrarily. Branch changes require persistent defect support, boundary support, or prescribed transition strata.

Axiom 14.7 (Variational selection). Admissible histories are selected by stationarity or local minimization of accumulated Kleban defect plus regularity, boundary, and branch-transition penalties.

15 Rules of the Calculus

Rule 15.1 (Latentization).

$$(S, T, \varepsilon) \mapsto \Lambda = \langle \text{Stab}_\varepsilon(S, T) \rangle / \sim.$$

Unstable substrate data are outside the effective domain.

Rule 15.2 (Branch projection).

$$\mathcal{R}_i^\beta = a_i e_i + J_i \Gamma_{\beta_i}(\rho_i^{\beta_i}, \theta + \alpha_i, \sigma_i).$$

No axis-specific formulas are primitive after J_i and Γ_b are defined.

Rule 15.3 (Kleban obstruction).

$$\mathbb{K}_{\theta, \beta} = \delta_A \mathcal{R}_\theta^\beta.$$

The Kleban Operator is the obstruction to mutual projection agreement.

Rule 15.4 (Equalizer).

$$\mathbb{K} = 0 \iff \mathcal{R}_0 = \mathcal{R}_1 = \mathcal{R}_2.$$

Exact realspace is the zero-defect equalizer.

Rule 15.5 (Tolerance).

$$\|\mathbb{K}\| \leq \varepsilon$$

defines finite-resolution realspace.

Rule 15.6 (Linearization).

$$D\mathbb{K} = \delta_A \circ d\mathcal{R}.$$

Infinitesimal defect is the coboundary of infinitesimal projection.

Rule 15.7 (Pullback geometry).

$$g_{\mathbb{K}}^N = (D\mathbb{K})^* \kappa(D\mathbb{K}).$$

This is normal defect geometry, not automatically intrinsic realspace geometry.

Rule 15.8 (Curvature).

$$\Omega_{\mathbb{K}} = \delta_A d(d\alpha + \omega).$$

Relative curvature is the coboundary of sector transport curvature.

Rule 15.9 (Gauge).

$$\theta \mapsto \theta + c, \quad \alpha_i \mapsto \alpha_i - c$$

preserves phase readout, and

$$\alpha_i \mapsto \alpha_i + \eta_i, \quad \omega_i \mapsto \omega_i - d\eta_i$$

preserves local gluing one-forms.

Rule 15.10 (Variation).

$$H^* \in \text{Crit} \sum_n \left(\|\mathbb{K}_n\|_{\kappa_n}^2 + \text{extregularity} + \text{boundary} + \text{branch variation} \right).$$

Histories are selected by constrained stationarity or local minimization.

16 Minimal Model

A model of Universal Calculus is a tuple

$$\mathfrak{U} = (I, S, T, \varepsilon, \Lambda, V, A, B, \theta, a, r, R, \alpha, \sigma, \beta, J, \mathcal{R}, \delta_A, \mathbb{K}, \mathcal{A}, \Omega_{\mathbb{K}}, \kappa, h, g_{\mathbb{K}}^N, g_{\mathbb{K}}^T, \mathcal{S}).$$

It satisfies the axioms above and therefore realizes

$$\mathfrak{U} \models \text{UC}_{\mathbb{K}}.$$

The compressed construction is

$$\begin{aligned} &\text{stable comparisons} + \text{one global phase} + \text{branched projection} \\ &\quad + \text{Kleban equalization} + \text{defect variation} \implies \text{local realspace.} \end{aligned}$$

17 Operator Grammar

Universal Calculus is the formal closure generated by the Kleban Operator under a small list of operations:

$$\text{UC}_{\mathbb{K}} = \left\langle \mathbb{K}; \ker, (\cdot)^{-1}(B_{\varepsilon}), d, *, \delta_A, \text{TV}, \text{Crit} \right\rangle.$$

Operation	Generated object
$\ker \mathbb{K}$	exact realspace compatibility locus
$\mathbb{K}^{-1}(B_{\varepsilon})$	finite-resolution compatibility locus
$d\mathbb{K}$	linearized defect operator
$(d\mathbb{K})^* \kappa(d\mathbb{K})$	defect-normal metric
$\delta_A d(d\alpha + \omega)$	relative gluing curvature
$\text{TV}(\beta)$	branch transition control
$\text{Crit}(\mathcal{S})$	admissible histories

18 Canonical Formula Sheet

Universal Calculus in one page

$$\begin{aligned}
\Lambda &= \langle \text{Stab}_\varepsilon(S, T) \rangle / \sim \\
\psi_i &= \theta + \alpha_i \\
\Gamma_b(\rho, \psi, \sigma) &= \begin{cases} \rho(\cos \psi, \sin \psi), & b = E, P, \\ \sigma \rho(\cosh \psi, \sinh \psi), & b = H, \end{cases} \\
q_b(u, v) &= u^2 + \varepsilon_b v^2, \quad q_b(\Gamma_b) = \rho^2 \\
\mathcal{R}_i^\beta &= a_i e_i + J_i \Gamma_{\beta_i}(\rho_i^{\beta_i}, \theta + \alpha_i, \sigma_i) \\
\mathbb{K}_{\theta, \beta} &= \delta_A \mathcal{R}_\theta^\beta \\
\text{Real}_0^\beta &= \ker \mathbb{K}_{\theta, \beta} = \text{Eq}(\mathcal{R}_0^\beta, \mathcal{R}_1^\beta, \mathcal{R}_2^\beta) \\
\text{Real}_\varepsilon^\beta &= \mathbb{K}_{\theta, \beta}^{-1}(B_\varepsilon(0)) \\
D\mathbb{K} &= \delta_A d\mathcal{R}_\theta^\beta \\
g_{\mathbb{K}}^N &= (D\mathbb{K})^* \kappa(D\mathbb{K}), \quad g_{\mathbb{K}}^T = (d\pi)^* h(d\pi) \\
\Omega_{\mathbb{K}} &= \delta_A d(d\alpha + \omega) \\
\mathcal{S}[H] &= \sum_n \left(\|\mathbb{K}_n\|_{\kappa_n}^2 + \lambda \mathcal{R}_n + \mu \mathcal{B}_n + \nu J_{\beta, n} \right), \quad H^* \in \text{Crit}(\mathcal{S}).
\end{aligned}$$

19 Known Unknowns

The calculus exposes specific classification problems.

Nondegeneracy

Classify

$$\ker D\mathbb{K}_\lambda.$$

Each null direction must be identified as gauge, physical tangent, or inadmissible degeneracy.

Equalizer existence

The set

$$\text{Eq}(\mathcal{R}_0^\beta, \mathcal{R}_1^\beta, \mathcal{R}_2^\beta)$$

may be empty, singular, disconnected, or branch-dependent. Existence and regularity require separate criteria.

Branch dynamics

The branch cost J_β is not unique. Different choices may produce different stable interfaces, different topology changes, and different defect concentrations.

Metric signature

Hyperbolic branches and indefinite contrast forms can generate pseudo-Riemannian signatures. A physical interpretation requires an independent signature-selection rule.

Observable correspondence

The formal variables θ , $\mathbb{K}$, κ , $g_{\mathbb{K}}$, and $\Omega_{\mathbb{K}}$ require correspondence maps before empirical predictions are defined.

20 Unknown Unknowns

The calculus also admits failure modes that are not ruled out by the formal definitions:

- singular branch caustics caused by repeated changes in β ;
- hidden nonlocal coupling induced by one global phase read through many local offsets;
- branch monodromy around persistent defect interfaces;
- chaotic minimizers of the action when branch cost competes with defect cost;
- topology change generated by stable finite- ε equalizers;
- degeneracy of the realspace image projection π^β ;
- mismatch between defect-normal stiffness and intrinsic tangential geometry;
- unstable real-form selection in the complexified target.

These are not contradictions. They are study targets for concrete models of $\text{UC}_{\mathbb{K}}$.

21 Terminal Statement

Definition of Universal Calculus

$$\text{UC}_{\mathbb{K}} = \left(\mathcal{L}_\varepsilon, \theta, B, \Gamma, J, \mathcal{R}, \delta_A, \mathbb{K}, \ker, d\mathbb{K}, \Omega_{\mathbb{K}}, \kappa, g_{\mathbb{K}}^N, g_{\mathbb{K}}^T, \mathcal{S} \right)$$

with primitive operator

$$\mathbb{K}_{\theta, \beta} = \delta_A \circ \mathcal{R}_\theta^\beta.$$

A local universe is a model of this calculus when

$$\begin{array}{lll} \mathbb{K} \approx 0, & \Omega_{\mathbb{K}} \text{ controlled,} & \text{TV}(\beta) < \infty, \\ H \in \text{Crit}(\mathcal{S}), & \text{chosen metric sector nondegenerate.} & \end{array}$$

Reference Envelope

The calculus is condensed from the manuscript *A Branched Tri-Cylindrical Construction of Local Realspace* by Nicholas Kleban, dated June 7, 2026. That manuscript defines the stable comparison substrate, one global phase, branch set $\{E, H, P\}$, axis reconstruction maps, realspace equalizer, gluing defect, curvature one-forms, pullback metric, branch admissibility, action principle, complex real forms, construction algorithm, and open problems.

The following references are included as structural anchors and analogues, not as claims that the present calculus is identical to any established framework.

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